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Project Title:	Class III Histone Deacetylases as a Defense Against Radiation-Induced DNA Damage		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
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	POSTDOCTORAL FELLOWSHIP. Space radiation presents a significant hazard to space flight crews. It is dominated by high energy charged particles such as protons and heavy ions, which may cause significant DNA damage at relevant exposure levels. Depending on the radiation dose, this DNA damage can range from an extent that is compatible with cell survival to levels that induce cell death. The latter can cause a potentially life-threatening decrease in radiation-sensitive cells such as white blood cells and platelets (anemia), the former may manifest itself in malignant transformation of irradiated cells, which can ultimately lead to cancer. In the past few years, a group of enzymes called the Sir2-class of histone deacetylases (class III HDACs) or "the sirtuins" have been shown to regulate cell defenses against DNA damage and biological stress. Sirtuins may further underlie the stress resistance and increased lifespan of mammals fed a calorie-restricted diet. In simple organisms such as yeast, worms and flies, additional copies of the Sir2 gene make cells more resistant to		

Task Description:	radiation and extend the organism's lifespan by ~30%. In rodents, calorie-restriction causes increased radiation resistance and a significantly lower cancer incidence, as well as lifespan extension. Yeast and fly data show that sirtuins are required for the maintenance of genomic integrity and directly aid the repair of DNA damage. While the precise role played by sirtuins in higher organisms remains elusive, published results demonstrate that upregulation of sirtuins leads to numerous cytoprotective events including increased cell survival and activation of DNA repair pathways. The goal of this proposal is to elucidate the mechanisms by which mammalian sirtuins may confer protection against DNA damage. To mimic increased sirtuin activity in animals, mouse models were be generated that allow for the inducible overexpression of individual sirtuin family members. The impact of increased sirtuin activity on radiation-induced damage will be analyzed with regard to both carcinogenesis and anemia. Importantly, sirtuin activity can be enhanced using sirtuin-activating compounds (STACs) that we have discovered, which makes sirtuins an ideal target for pharmacological protection from radiation-induced pathologies during or follwing space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In the past few years, a group of enzymes called "sirtuins" have been shown to regulate cell defenses against damage and biological stress in diverse organisms, from yeast to mammals. They are thought to have evolved to help organisms survive adversity and may underlie the stress resistance and increased lifespan of mammals fed a calorie-restricted diet. In rodents, caloric restriction is one of the most effective ways to protect animals from radiation-induced DNA damage and cancer. Indeed, our lab has shown that calorie-restricted rats have greatly elevated levels of the mouse sirtuin SIRT1 and that having additional copies of SIRT1 protects mammalian cells from radiation. In simple organisms such as yeast, worms and flies, additional copies of the sirtuin gene SIR2 gene make cells more resistant to radiation and extend average lifespan by ~30%. Thus, harnessing sirtuins may provide a means to greatly increase the survival of critical cells such as stem cells, white blood cells and neurons in response to radiation damage and possibly prevent radiation-induced cancers. Sirtuins can be activated using sirtuin-activating components (STACs) such as resveratrol, a small polyphenolic molecule produced by stressed plants. We further demonstrated that resveratrol promotes the survival of fibroblasts in response to irradiation. My current research corroborates these findings and places SIRT1 in the context of DNA damage repair upon exposure to free radicals, a common and deleterious by-product of ionizing radiation. Radiation exposure is not limited to astronauts and our research may, therefore, provide benefits for professionals facing elevated radiation exposure levels such as airline personnel and radiation workers. Our research may further have important medical benefits for a large number of the of the US population. Free radicals are generated as part of daily living, as a consequence of respiration and energy metabolism. Free radical-induced damage is a major contributor to age-related organ decline and cancer. Our research explores the consequences of increased SIRT1 activity on free radical-induced damage. Specifically, it has been demonstrated recently that oxidative stress alters gene expression in the aging human brain. Deregulation of gene families involved in synaptic plasticity and other key neurobiological pathways was found to increase with age and correlate with oxidative stress. Our research may be exploited to slow down the aging process by mitigating the consequences of oxidative stress on the human body. Indeed, we have shown that increased SIRT1 activity can oppose gene expression changes induced by free radicals. We envisage that STACs may find broad use in the US as a pharmacological agent against cancer, the age-related decline in brain function, and possibly aging itself.
Task Progress:	The goal of this study is to investigate the potential role of class III histone deacetylases, also known as ‘sirtuins’, in the maintenance of genomic stability and in the cellular defense against radiation damage. Given the deleterious effect of space radiation on genomic DNA, this proposal will provide useful insights into molecular mechanisms that may be utilized to protect space flight crews from the potentially detrimental consequences of exposure to space radiation. One of the deleterious consequences of space irradiation is the generation of reactive oxygen species (ROS) and their damaging impact on both cellular proteins and DNA. The latter can cause genomic instability and, ultimately, cancer. Our lab works on SIRT1-activating compounds (STACs) that might one day be taken by astronauts to protect against radiation damage. Our recent data suggest a conserved role for sirtuins in the maintenance of genomic integrity and protection from DNA damage, in organisms as diverse as yeast and mammalian cells. Specifically we found that, in yeast, oxidative stress results in a loss of silencing and increased DNA instability at loci that are normally repressed by the sirtuin called "Sir2". Paralleling yeast Sir2, its mouse orthologue SIRT1 was shown to participate in the silencing of repetitive (pericentromeric) DNA and this silencing is reduced upon exposure to ROS. By mapping the promoters that SIRT1 binds to, we further detect a major redistribution of SIRT1 upon exposure to DNA damage that correlates with alterations in the expression of SIRT1-target genes. Preliminary evidence suggests that SIRT1 is recruited to sites of ROS-damaged DNA, possibly to facilitate repair. Indeed, SIRT1-deficient embryonic stem cells are more susceptible to ROS-induced genomic instability. These findings are corroborated by biochemical analyses demonstrating that the total amount of chromatin-associated SIRT1 is dramatically increased upon ROS exposure as well as in response to irradiation. Together our data indicate that the genomic localization of SIRT1 may change during oxidative stress and thereby affect DNA damage repair and/or the maintenance of genomic integrity upon ROS exposure. To address whether these findings have an implication for radiation-induced pathologies such as cancer and anemia, we generated a mouse model that allows for temporally or spatially regulated SIRT1 overexpression. These mice were then crossed to cancer models of genomic instability that are highly susceptible to radiation-induced DNA damage. We are currently following a cohort of experimental and control mice and will be able to determine the effect of increased SIRT1 activity on genomic stability and cancer induction and/or progression in response to radiation within the next months. In addition to transgenic sirtuin overexpression, sirtuin activating small molecules (STACs) will be tested in the same cancer-prone mouse models.
Bibliography Type:	Description: (Last Updated: 06/09/2008)
Abstracts for Journals and Proceedings	Firestein R, Michan S, Oberdoerffer P, Campbell J, Luikenhuis S, deCabo R, Sinclair DA. "SIRT1 is a tumor suppressor in vivo." Molecular Genetics of Aging, Cold Spring Harbor Laboratory, October 2006. Molecular Genetics of Aging, Cold Spring Harbor Laboratory, October 2006. , Oct-2006